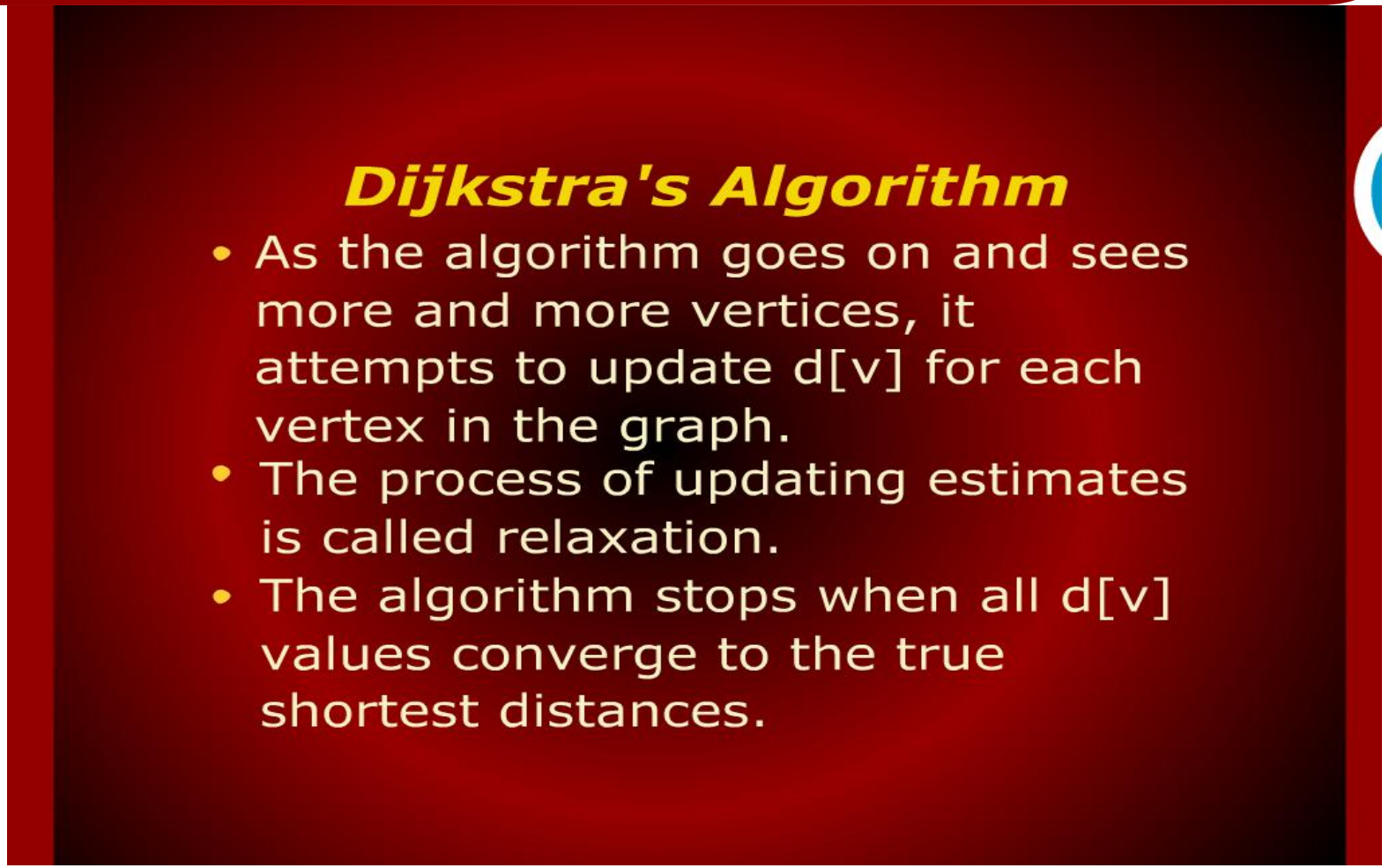
Algorithms

Lecture #39

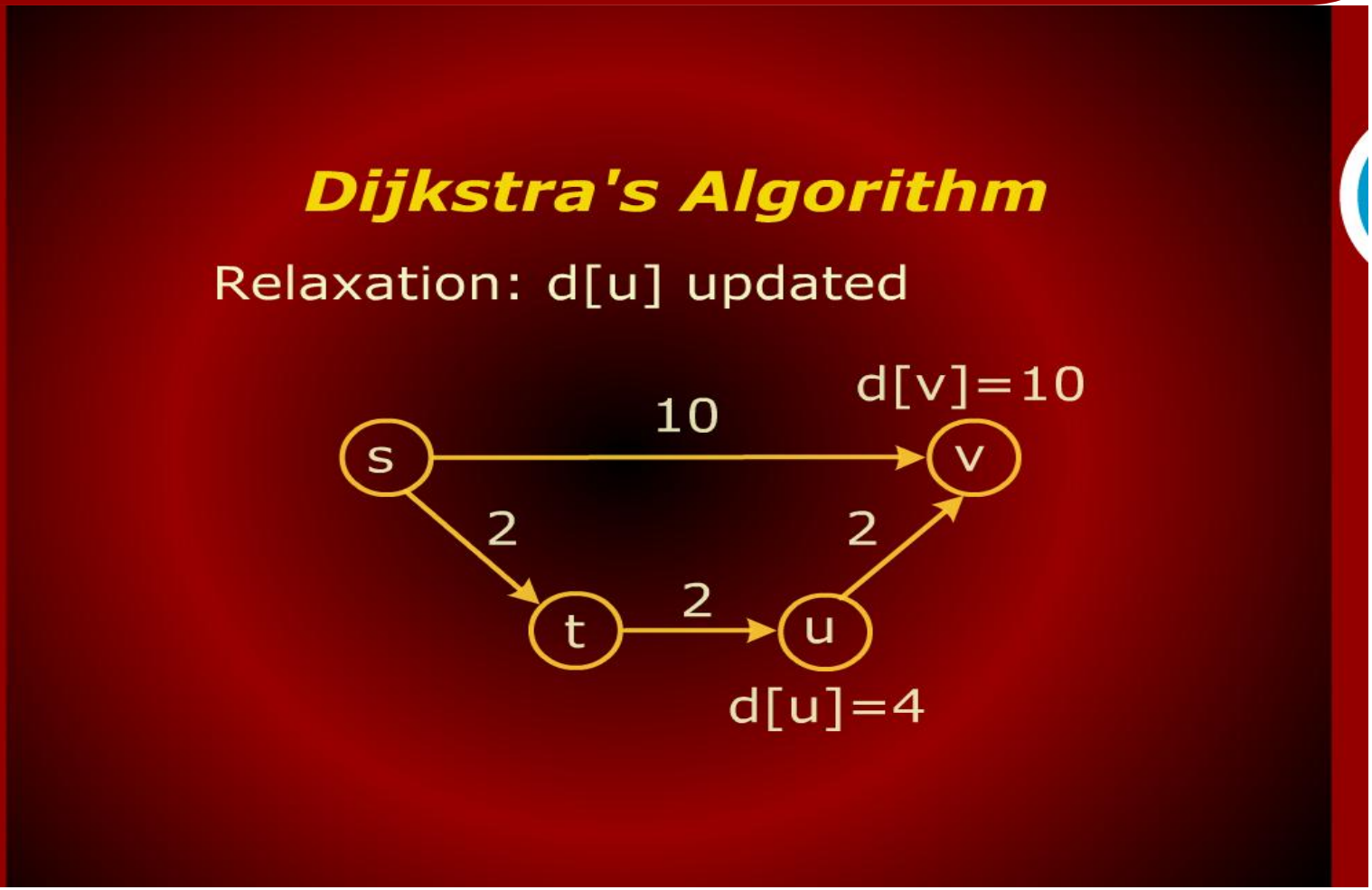
Syed Hamed Raza

Dijkstra’s Algorithm

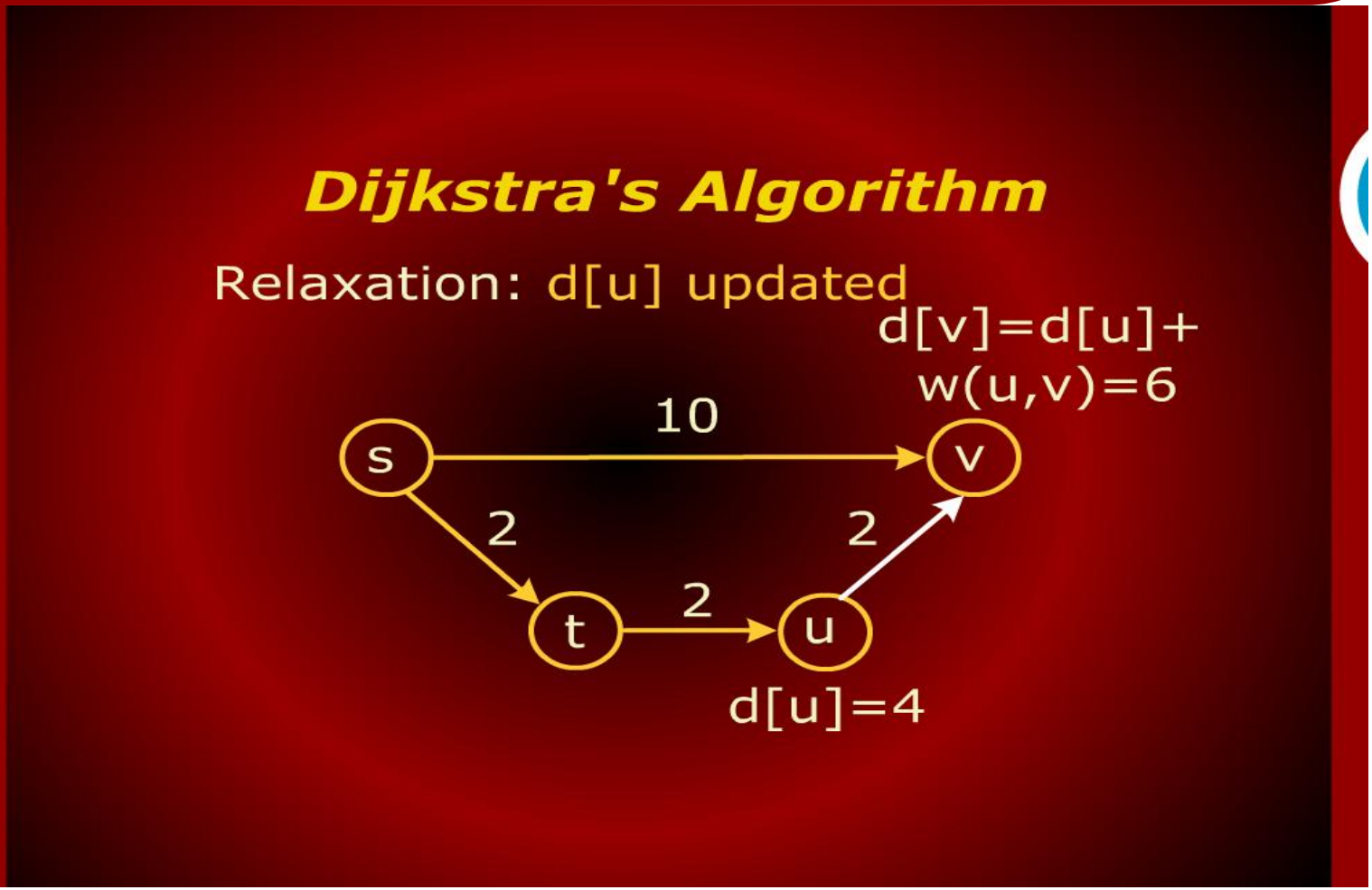
Dijkstra’s Algorithm



Dijkstra’s Algorithm



Dijkstra's Algorithm



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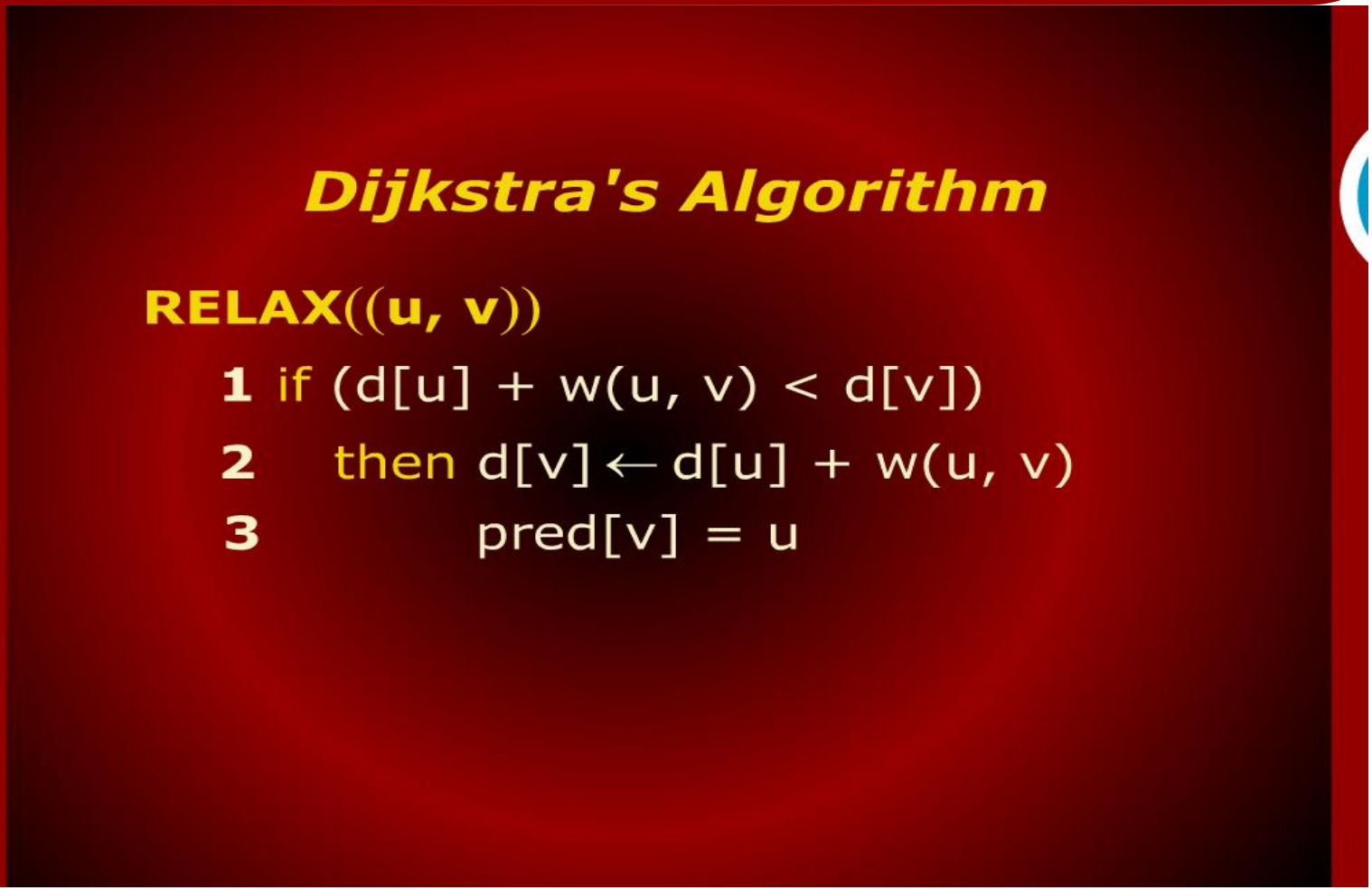


Dijkstra's Algorithm

Dijkstra's Algorithm

- Dijkstra's algorithm is based on the notion of performing repeated relaxations.
- Dijkstra's algorithm operates by maintaining a subset of vertices, $S \subseteq V$, for which we claim we know the true distance,
- $d[v] = \delta(s, v)$.

Dijkstra's Algorithm

Dijkstra’s Algorithm

Dijkstra's Algorithm

- How do we select which vertex among the vertices of V–S to add next to S?
- Here is greediness comes in.
- For each vertex u $\in$ (V–S), we have computed a distance estimate d[u].

Dijkstra's Algorithm

Dijkstra's Algorithm

- The greedy thing to do is to take the vertex for which d[u] is minimum,
- i.e., take the unprocessed vertex that is closest by our estimate to s.
- Later, we justify why this is the proper choice.
- In order to perform this selection efficiently, we store the vertices of V−S in a priority queue.

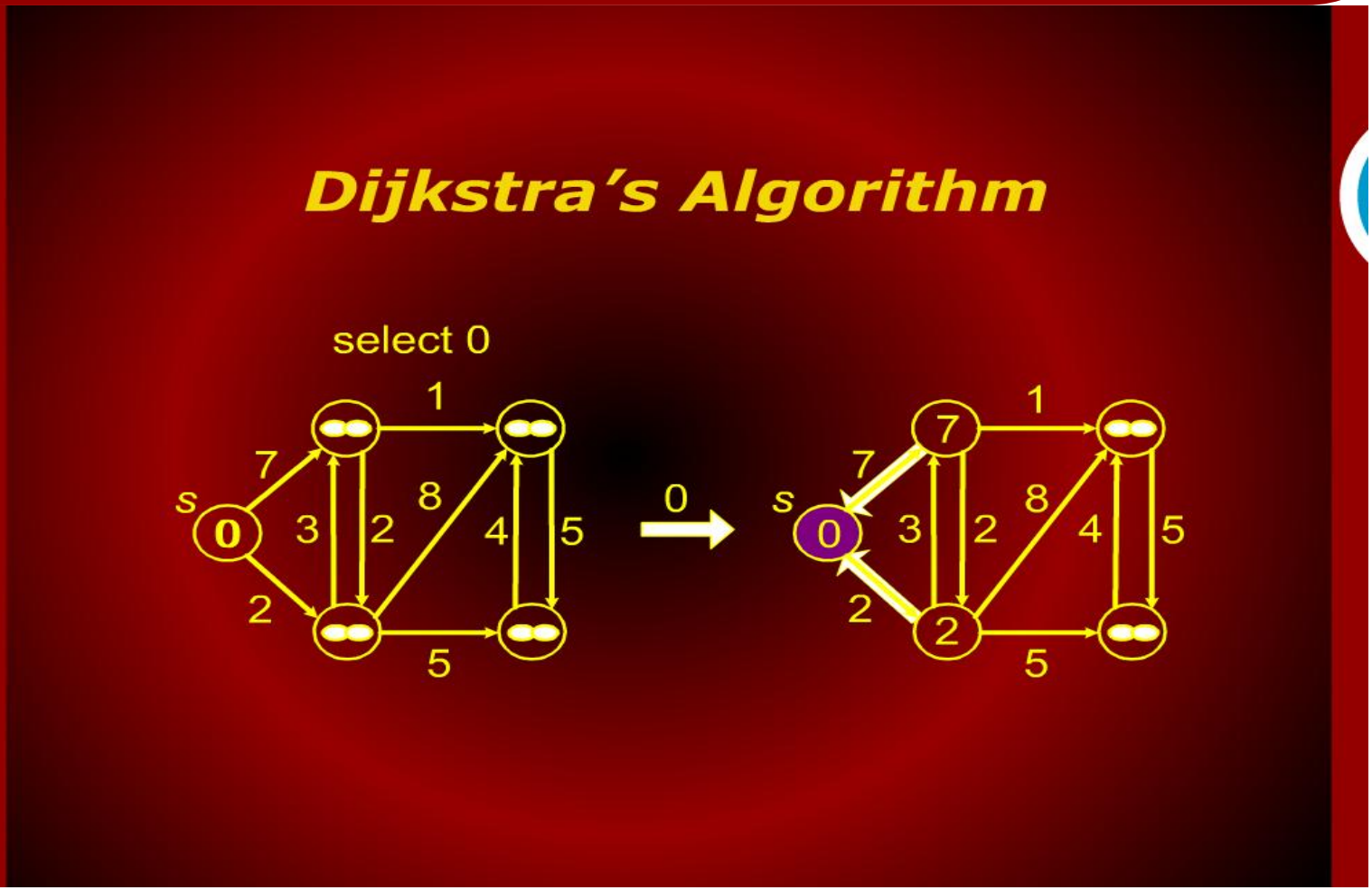
Dijkstra's Algorithm

Dijkstra's Algorithm

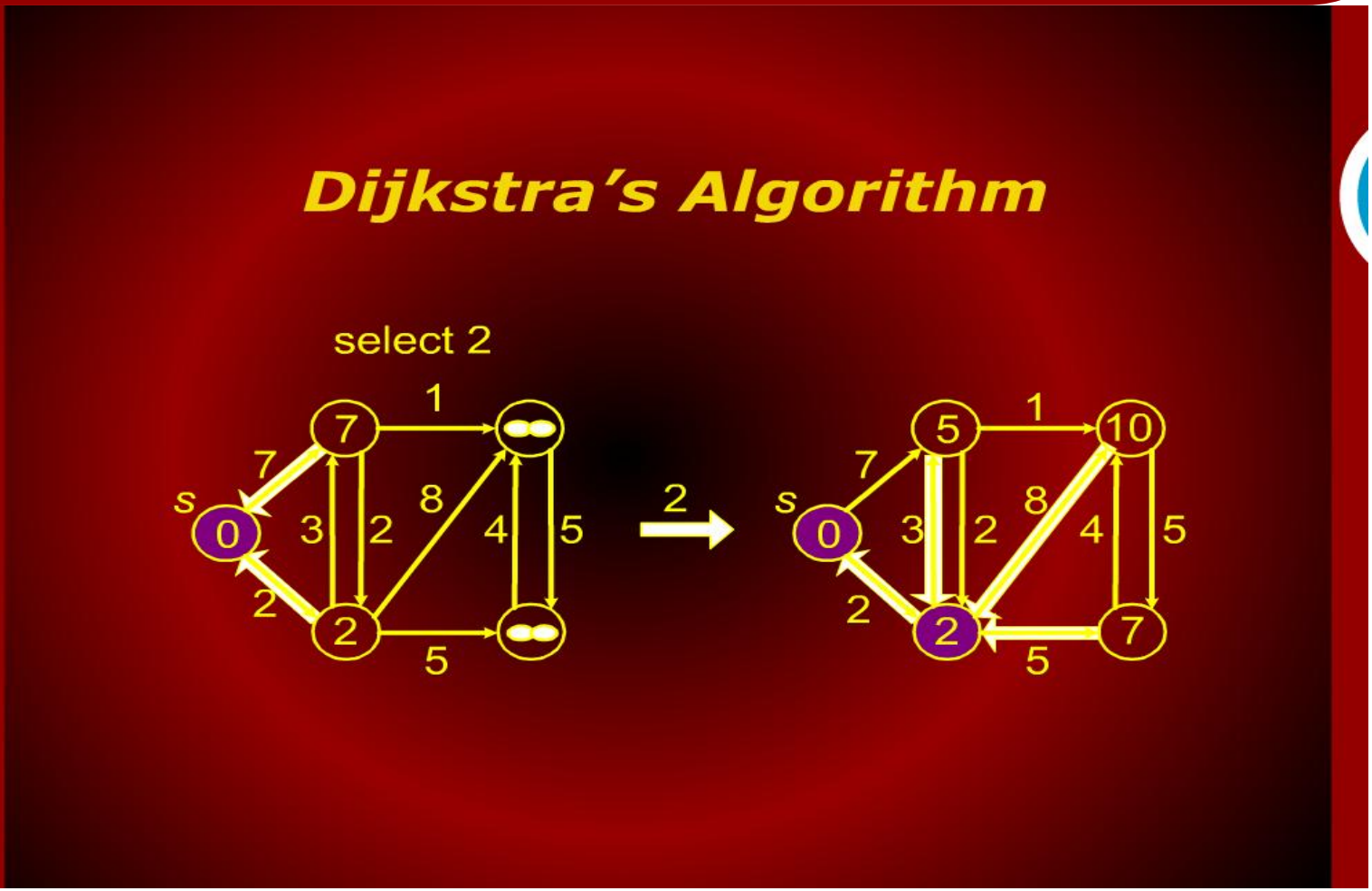
Dijkstra's Algorithm

- Note the similarity with Prim's algorithm, although a different key is used here.
- Therefore the running time is the same, i.e., $\Theta(E \log V)$.

Dijkstra's Algorithm



Dijkstra's Algorithm



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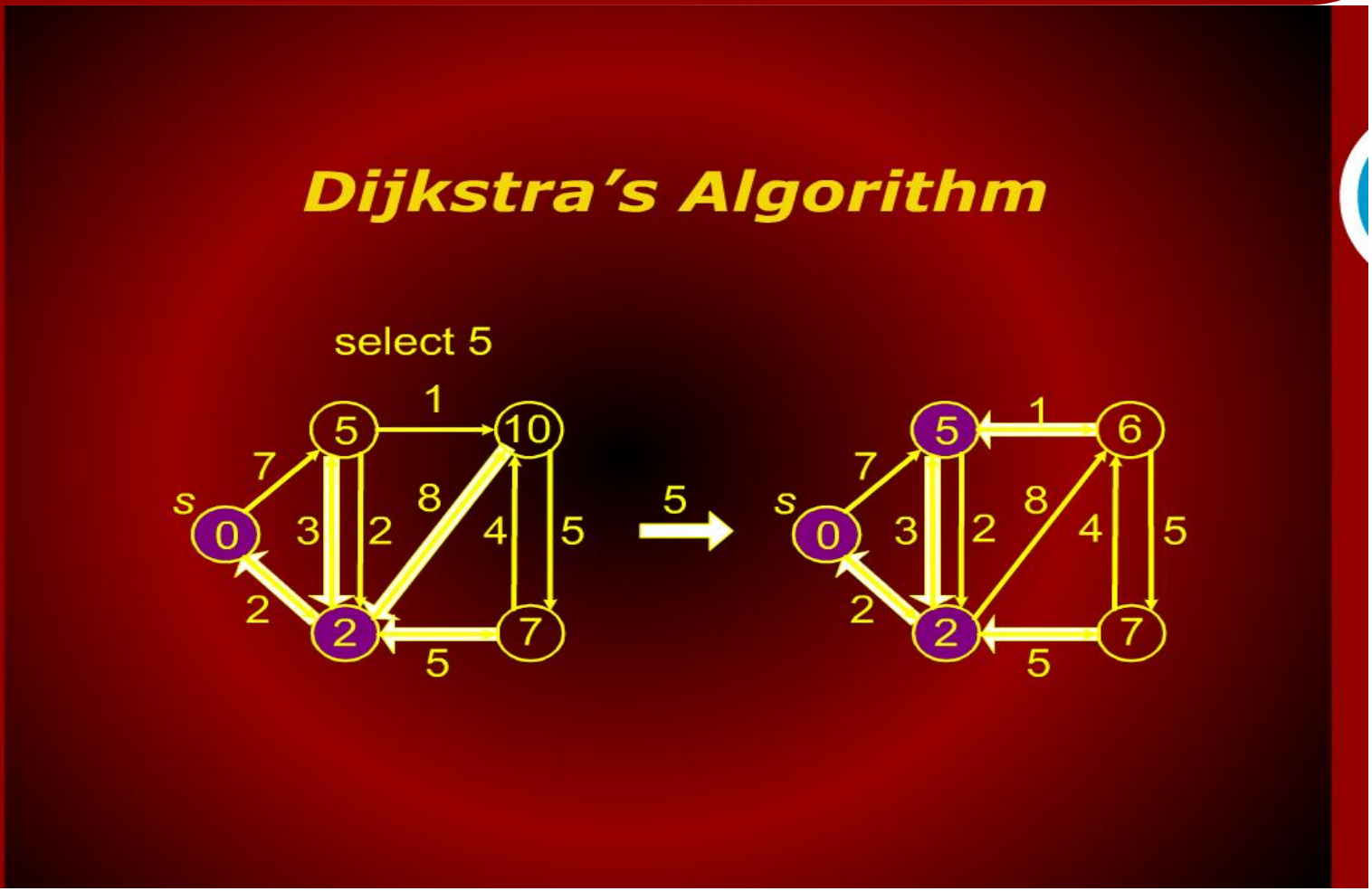

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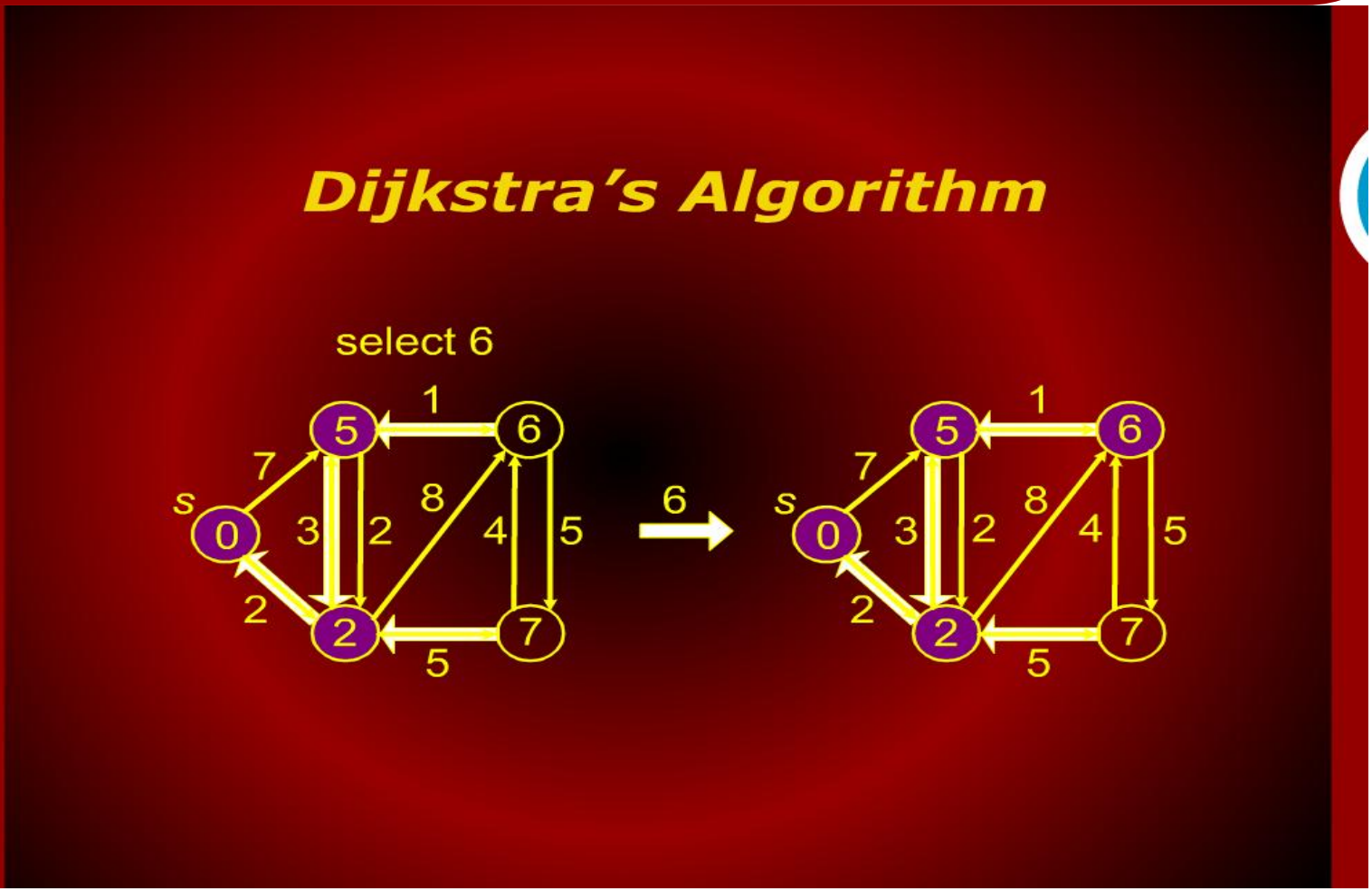

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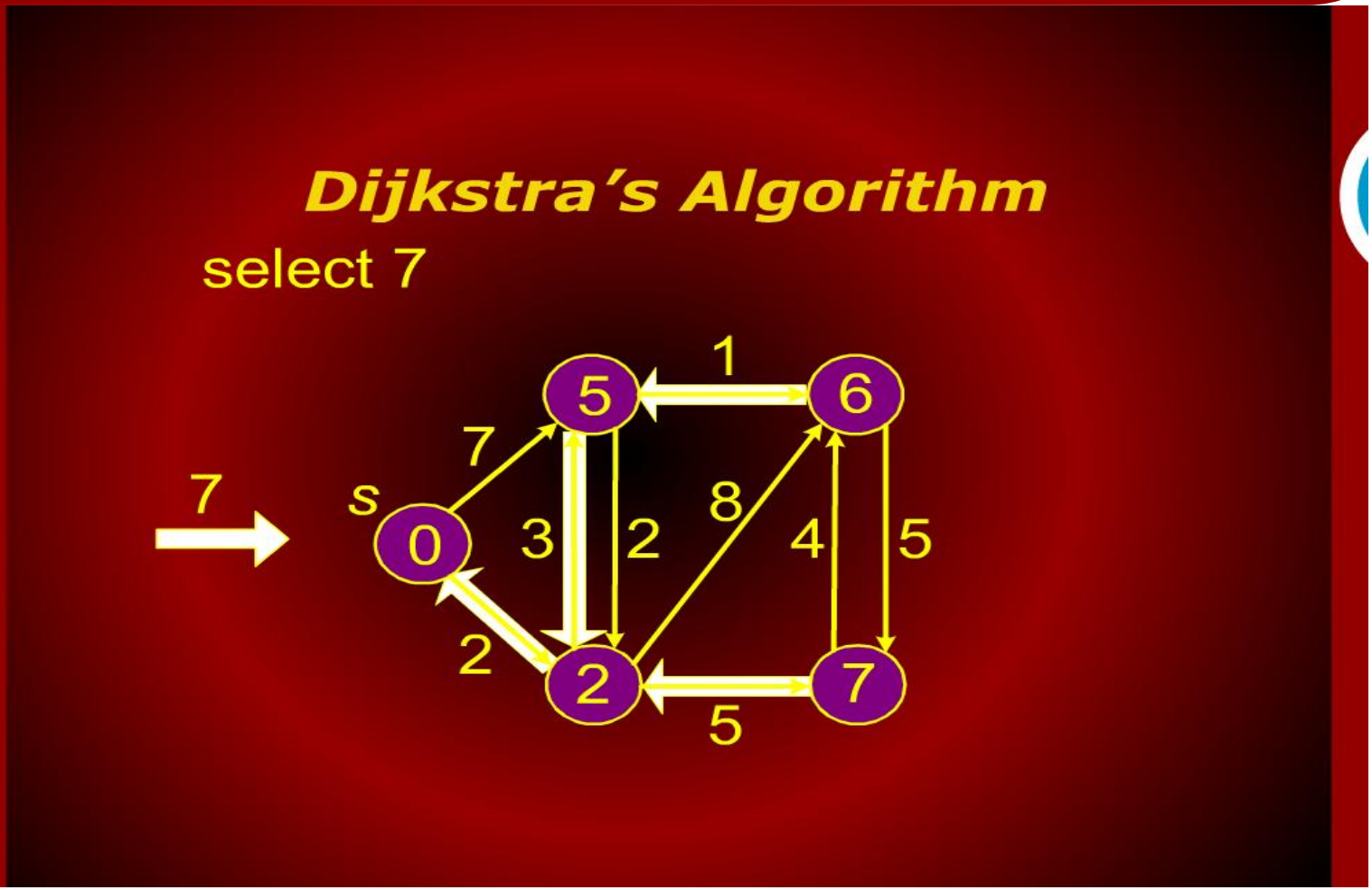
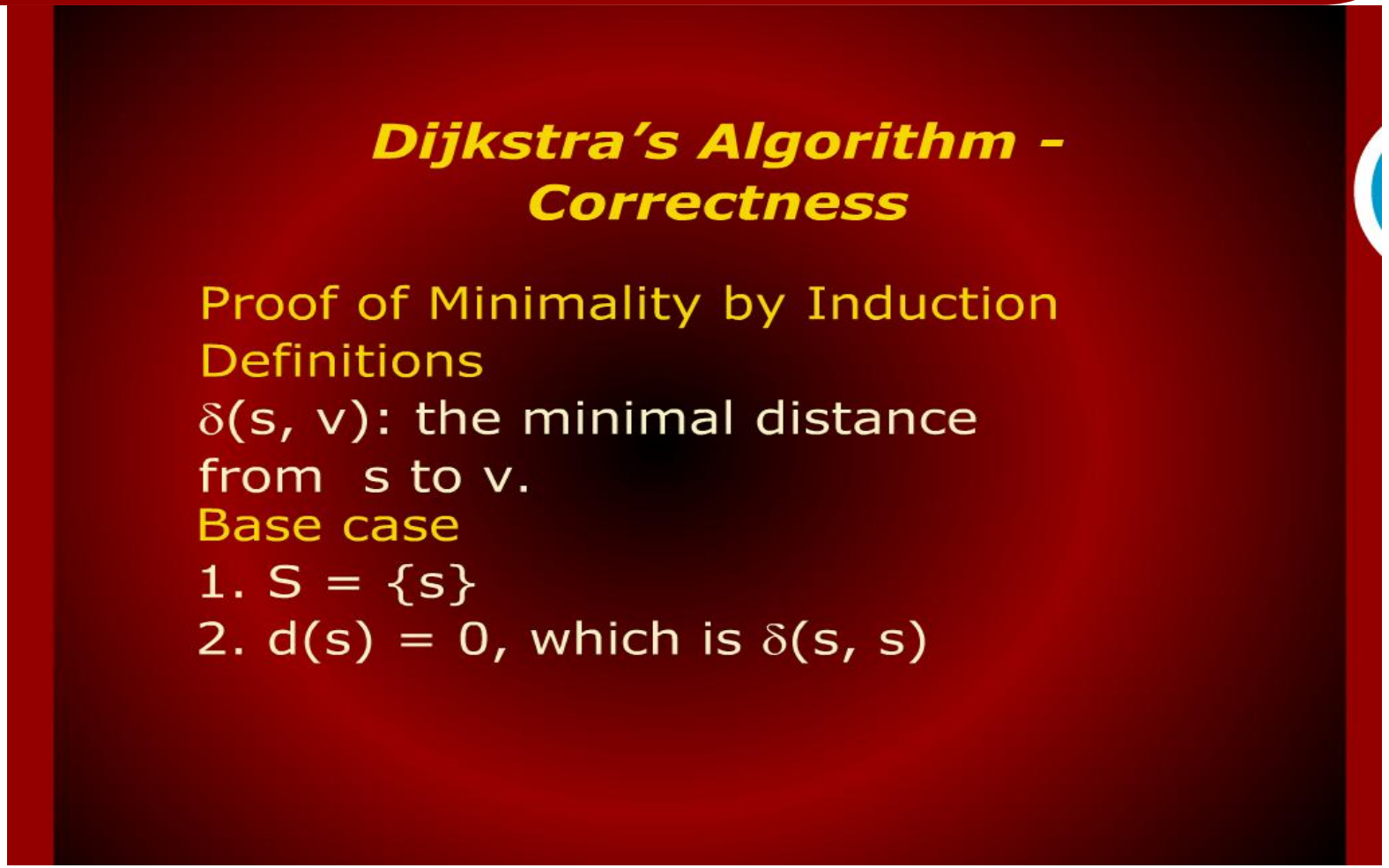


Dijkstra's Algorithm – Correctness



Dijkstra's Algorithm – Correctness

Dijkstra's Algorithm
- Correctness

Assume

1. $d(v) = \delta(s, v)$ for every $v \in S$
2. All neighbors of v have been relaxed.

Dijkstra's Algorithm – Correctness

Dijkstra's Algorithm - Correctness

Proof

- If $d(u) \leq d(u')$ for every $u' \in V$ then $d(u) = \delta(s, u)$, and we can transfer u from V to S, after which $d(v) = \delta(s, v)$ for every $v \in S$.
- We do this as a proof by contradiction.

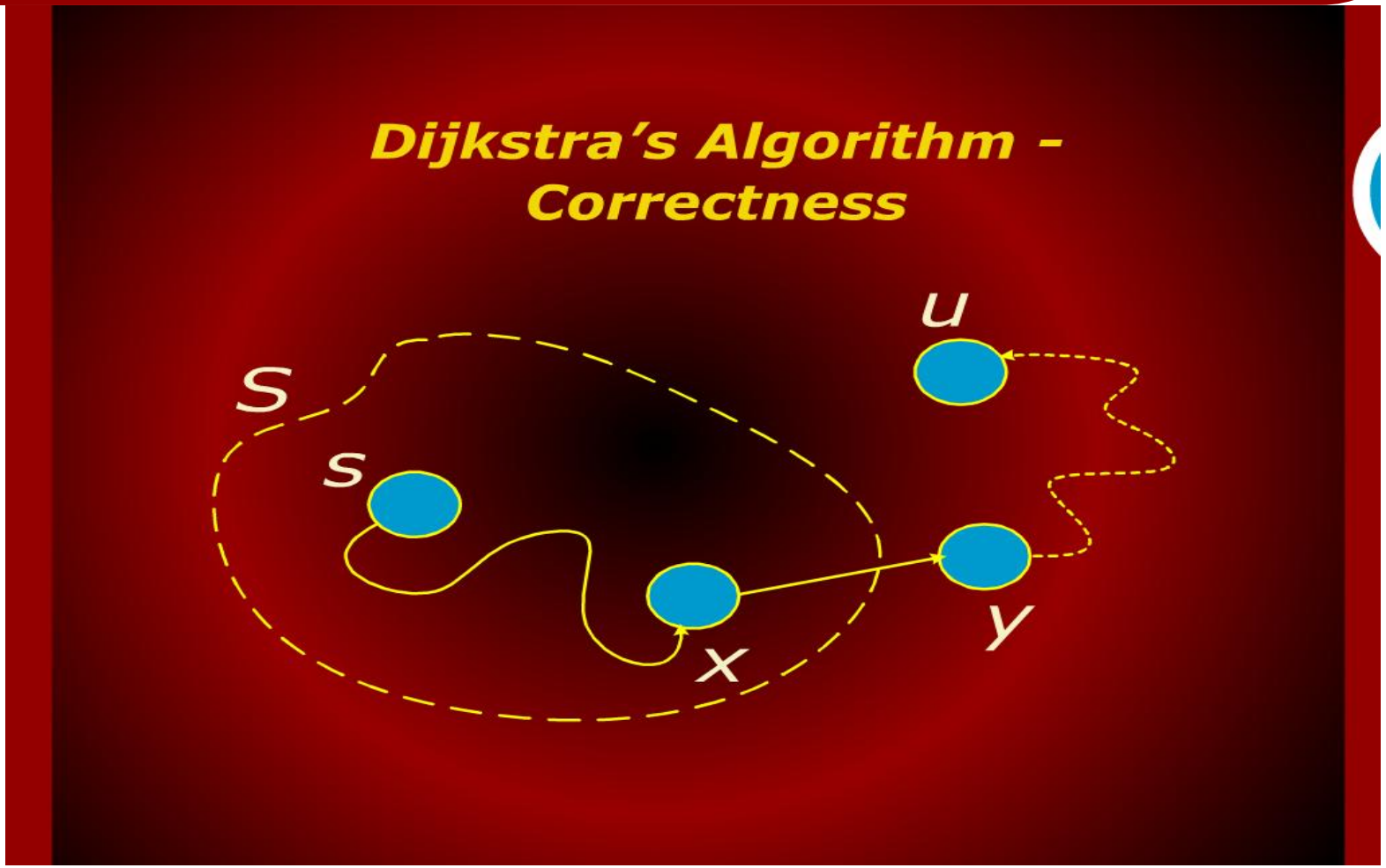
Dijkstra's Algorithm – Correctness

Dijkstra's Algorithm - Correctness

Suppose that $d(u) > \delta(s, u)$.

- The shortest path from s to u, p(s, u), must pass through one or more vertices exterior to S.
- Let x be the last vertex inside S and y be the first vertex outside S on this path to u.
- Then $p(s, u) = p(s, x) \cup \{(x, y)\} \cup p(y, u)$.

Dijkstra's Algorithm – Correctness



Kruskal's Algorithm

Kruskal's Algorithm

- The length of $p(s, u)$ is $\delta(s, u) = \delta(s, y) + \delta(y, u)$.
- Because y was relaxed when x was put into S, $d(y) = \delta(s, y)$ by the convergence property.
- Thus $d(y) \leq \delta(s, u) \leq d(u)$.
- But, because d(u) is the smallest among vertices not in S, $d(u) \leq d(y)$.
- The only possibility is $d(u) = d(y)$, which requires $d(u) = \delta(s, u)$ contradicting the assumption.